

```
import google.generativeai as genai
from dataclasses import dataclass
from langgraph.graph import StateGraph, END
import os
import pickle
```

```
import google.generativeai as genai

GEMINI_API_KEY = "AIzaSyBtStk2YSGVkdI_gUij8i4llXygwLe_Dtw"
genai.configure(api_key=GEMINI_API_KEY)
```

```
def call_llm(prompt: str) -> str:
    model = genai.GenerativeModel("gemini-2.5-flash")
    response = model.generate_content(prompt, generation_config={"temperature":0.5})
    return response.text
```

```
def generate_post(topic, platform):
    prompt = f"Generate an engaging social media post for {platform} about: {topic}"
    return {"post": call_llm(prompt)}
```

```
def generate_hashtags(post_text):
    prompt = f"Generate relevant hashtags for this post:\n{post_text}"
    return {"hashtags": call_llm(prompt)}
```

```
def reply_to_comment(comment):
    prompt = f"Write a polite and engaging reply to this comment:\n{comment}"
    return {"reply": call_llm(prompt)}
```

```
# LangGraph state
@dataclass
class SocialMediaState:
    topic: str = ""
    platform: str = ""
    post: str = ""
    hashtags: str = ""
```

```
comment: str = ""
reply: str = ""
```

```
# Build LangGraph workflow
def build_social_graph():
    graph = StateGraph(SocialMediaState)

    graph.add_node("generate_post", lambda s: generate_post(s.topic, s.platform))
    graph.add_node("generate_hashtags", lambda s: generate_hashtags(s.post))
    graph.add_node("reply_comment", lambda s: reply_to_comment(s.comment))

    graph.set_entry_point("generate_post")
    graph.add_edge("generate_post", "generate_hashtags")
    graph.add_edge("generate_hashtags", "reply_comment")
    graph.add_edge("reply_comment", END)

    return graph.compile()
```

```
# Instantiate agent
social_agent = build_social_graph()
```

```
# Run function
def run_social_agent(topic, platform="Twitter", comment=""):
    state = SocialMediaState(topic=topic, platform=platform, comment=comment)
    result = social_agent.invoke(state)
    return {
        "post": result.get("post"),
        "hashtags": result.get("hashtags"),
        "reply": result.get("reply")
    }
```

```
output = run_social_agent(
    topic="AI tools for small businesses",
    platform="LinkedIn",
    comment="This is very insightful!"
)
print(output)
```

```

# List of sample social media tasks
questions = [
    {"topic": "AI tools for small businesses", "platform": "LinkedIn", "comment": "Very insightful!"},
    {"topic": "Sustainable fashion trends", "platform": "Instagram", "comment": "Love this!"},
    {"topic": "Top productivity apps in 2025", "platform": "Twitter", "comment": "Great suggestions!"},
    {"topic": "Healthy meal prep tips", "platform": "Facebook", "comment": "Thanks for sharing!"},
    {"topic": "Latest advancements in AI agents", "platform": "LinkedIn", "comment": "Impressive insights!"},
    {"topic": "How to grow your YouTube channel", "platform": "YouTube", "comment": "Amazing tips!"},
    {"topic": "Top 10 coding best practices", "platform": "Twitter", "comment": "Very helpful!"},
    {"topic": "Remote work productivity hacks", "platform": "LinkedIn", "comment": "Useful info!"},
    {"topic": "Eco-friendly travel destinations", "platform": "Instagram", "comment": "Great ideas!"},
    {"topic": "Tips for beginner photographers", "platform": "Facebook", "comment": "Nice suggestions!"}
]

# Loop over each prompt and run the agent
for q in questions:
    print(f"\n=== Topic: {q['topic']} | Platform: {q['platform']} ===\n")
    output = run_social_agent(topic=q['topic'], platform=q['platform'], comment=q['comment'])
    print("Generated Post:\n", output['post'])
    print("Hashtags:\n", output['hashtags'])
    print("Reply to Comment:\n", output['reply'])
    print("\n" + "-"*80)

```


=== Topic: AI tools for small businesses | Platform: LinkedIn ===

Generated Post:

Here are a few options for an engaging LinkedIn post about AI tools for small businesses, choose the one that best fits your tone!

****Option 1: The "Empowerment & Efficiency" Angle****

 ****Small Business Owners, Ready to Level Up Without Breaking the Bank?**** 

AI isn't just for tech giants anymore. It's becoming the ultimate equalizer, empowering small businesses to operate with incredible efficiency.

Imagine:

- * ****Automating customer support**** with intelligent chatbots, freeing up your team.
- * ****Generating compelling marketing copy**** for social media, emails, and blogs in minutes.
- * ****Analyzing sales data**** to spot trends and personalize customer experiences effortlessly.
- * ****Streamlining administrative tasks**** like scheduling and email responses.

These tools aren't replacing human ingenuity; they're amplifying it, allowing you to focus on what you do best: growing your business and serving your customers.

****What's one task in your small business you wish AI could handle? Share below! 🙋****

#AITools #SmallBusiness #Entrepreneurship #BusinessGrowth #Productivity #Innovation #SMB

****Option 2: The "Problem/Solution & Myth-Busting" Angle****

 ****Tired of wearing ALL the hats in your small business? There's a powerful co-pilot waiting.**** 

Many small business owners think AI is too complex, too expensive, or just "not for them." Think again!

Today's AI tools are intuitive, affordable, and designed to tackle common small business challenges head-on:

- * ****Content Creation:**** Stuck on your next blog post or social media caption? AI can draft ideas and even full articles.
- * ****Customer Service:**** Provide 24/7 support without hiring more staff, answering FAQs instantly.
- * ****Marketing Personalization:**** Deliver targeted messages that resonate, improving conversion rates.
- * ****Data Insights:**** Understand your customers and market better, without needing a data scientist.

It's about working smarter, not harder. AI can be your virtual assistant, marketing expert, and data analyst all rolled into one.

****If you haven't explored AI for your business yet, what's holding you back? Let's discuss!****

#SmallBusinessSuccess #AIforBusiness #DigitalTransformation #WorkSmarter #EntrepreneurLife #TechForSMBs
